

# An Area-Efficient Lookup-Table -Based eDRAM Digital CIM Macro for Neural Network Inference

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**Abstract**—Digital circuit is a promising approach to implement computing-in-memory (CIM) architecture for data-intensive applications, such as neural network inference. Previous digital CIM implementations have demonstrated higher throughput and robustness than the analog counterpart, benefiting from technology scaling and full-precision datapaths. However, the traditional digital CIM designs adopt a bulky adder tree as the computing circuit, limiting further improvements in area and energy efficiency. To address the above limitation, this work proposes a novel design methodology that combines the lookup-table (LUT) and high-density embedded dynamic RAM (eDRAM) cells. By storing precomputed weight summations in LUTs, the proposed approach reduces the first two stages of the adder tree. Additionally, the row-level parallelism of the CIM architecture is utilized to amortize the overheads of LUT weight encoding and eDRAM refreshing. The 28-nm prototype demonstrates a peak area efficiency of 16.2 TOPS/mm<sup>2</sup> and an energy efficiency of 49.5 TOPS/W, both for 8-bit operations. The proposed eDRAM macro achieves a 2.4 and 0.48 Mb/mm<sup>2</sup> density in memory and CIM mode, due to its balanced computing-storage ratio, enabling reconfigurability between computing cores and memory storage.

**Index Terms**—Computing-in-memory (CIM), digital circuit, embedded dynamic RAM (eDRAM), lookup-table (LUT).

## I. INTRODUCTION

**M**ATRIX-VECTOR multiplication lies in the heart of modern deep neural network models, where the huge weight parameters and massive multiply-and-accumulation

(MAC) operations put severe challenges for traditional computing platforms, such as CPU and GPU [1], [2]. The fundamental problem is due to the inefficient data transmission between memory array and computing units for classic Von Neumann architectures [3], [4]. Therefore, the computing-in-memory (CIM) architecture is proposed to minimize the overhead by equipping memory array with computing capability [5], [6]. Such approach enables both parallel readout and in situ dimension reduction, improving the throughput and efficiency at the same time [7], [8], [9].

The proposal of digital CIM [10], [11] marks a milestone in the history of CIM developments. The digital approach directly integrates standard logic cells aside the bit cell. Compared with previous analog implementations [8], [12], [13], [14], the standard logic cell guarantees the robustness for reliable neural network inference. Also, the digital circuits can benefit more from the advanced technology with increased transistor density and lower operating voltage, which brings both higher area and energy efficiency simultaneously [15], [16], [17]. Most importantly, the digital CIM macro provides easier integration with the digital SoC, especially when scaling up to large-scale CIM systems [18]. Actually, the developments of digital CIM have led to the widespread industrial application of CIM technology, ranging from embedded memories to DRAM [19], [20], [21].

Although previous digital CIM implementations have demonstrated the advantages of digital CIM in terms of accuracy, performance, and scalability, the efficiency is fundamentally limited by the bulky digital logic [22]. As shown in Fig. 1, a typical digital CIM macro can be transformed into a traditional weight stationary PE array by replacing the memory bit cells with latches. Considering that both adopt the same digital computing circuits, the main difference lies in the storage devices. However, the area of a fully parallel digital CIM array is mostly occupied by the massive logic cells, making the choice of memory cells less significant. As a result, previous digital CIM macros [11], [15], [16], [17], due to its limited storage density, behave more like a computing engine, instead of a memory array. This leads us to an important question, if the benefit of digital CIM solely comes from customized logic cells and manual layout?

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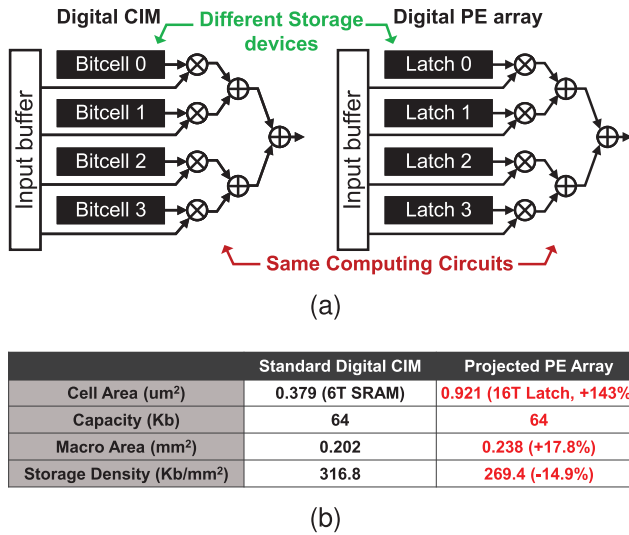


Fig. 1. (a) Illustration of digital CIM and traditional weight-stationary digital PE array. (b) Comparison of a 22-nm digital CIM macro [11] and the PE array projected by replacing SRAM with latches.

Sun et al. [23] summarize the mainstream digital CIM optimization approaches ranging from customized full adder to algorithm co-design. Although improving energy efficiency significantly, most approaches can also be applied to a traditional digital design flow by merging the customized cells into standard cell libraries. In other words, both the efficiency of digital CIM and digital accelerators are still fundamentally limited by digital circuits.

In this work, we break through the aforementioned limitation by revisiting the digital CIM design methodology. Rather than directly accumulating the results via full adders, we encode the weight summations into an area-efficient lookup-table (LUT) made up of high-density embedded dynamic RAM (eDRAM) cells. This allows the MAC operations to be replaced by efficient memory reads, thereby eliminating the first two stages of the adder tree. The proposed eDRAM-LUT design leverages two features of CIM architecture simultaneously: stationary weights and high-density memory cells. This distinguishes our design from previous digital CIM work that still relied on the traditional digital circuit design methodology. The main contributions of our work are listed as follows.

- 1) We propose an area-efficient eDRAM-LUT-based adder (eLAD) tree, which reduces 42% area and 58% energy than a conventional digital CIM design.
- 2) We reduce the overhead of LUT encoding and eDRAM refreshing by  $2.9\times$  and  $16\times$  via an in-memory acceleration approach.
- 3) We propose a high-density computing-storage reconfigurable eDRAM CIM macro with a density FoM<sup>1</sup>  $5.8\times$  higher than previous designs.

This article is an extended version of [25]. Compared to the previous conference version, we split the voltage supplies of computing circuits and eDRAM write drivers to achieve lower operating voltage and  $2.5\times$  higher energy efficiency.

<sup>1</sup>The density FoM is defined as the product of storage density and 8-bit MAC operation area efficiency [24].

The eDRAM cell is also optimized with an improved biasing scheme to extend the retention time by  $7.7\times$ . The layout of other computing and peripheral circuits is also optimized to reduce the whole macro area by 5.5%.

## II. BACKGROUND AND CHALLENGES

### A. Design Motivations

While the LUT-based computing technique has been explored since the emergence of CNN accelerators [26], [27], it was first adopted in digital CIM by [28], where a logic-based LUT was implemented to reduce dynamic power. Compared to the conventional digital CIM design, the logic-based LUT replaces the first stage of the adder tree with a 4-to-1 multiplexer. By using a static adder to compute the weight summation, the total dynamic power is reduced by 21%. While such approach does utilize the weight stationary property of the CIM architecture, its digital logic area is actually increased due to the additional logic, which requires more input bit parallelism for amortization. The proposed eDRAM-LUT design moves one step further by building a memory-based LUT design. The area overhead of LUT can be fully tolerated by the high-density memory cells, which allows us to exploit more aggressive adder tree reduction. Compared to the both previous LUT-based CIM and traditional digital accelerators [26], [27], [29], which still rely on standard cells to build LUT, the proposed eDRAM-LUT design is both area and energy efficient for general MAC operations.

On the other hand, our eDRAM-LUT-based digital CIM architecture is also motivated by previous reconfigurable analog CIM designs [30], [31]. The capacitors from the charge-based CIM array can be reused as the CDAC in the SAR-ADC for area reduction. Similarly, the CIM macro built by high-density eDRAM LUTs can be reconfigured as a normal memory buffer to reduce the data transfer between different macros. The effectiveness of the dual-mode CIM has been demonstrated in previous work with reconfigurable dataflow [24] and optimized mapping strategy [32]. Compared to previous analog reconfigurable designs that rely on a multiplexer to switch between ADC and CIM modes, the eDRAM-LUT design enables reconfiguration by simply updating the LUT content, incurring minimal area overhead while fully utilizing all bit-cells.

### B. Challenges to Efficient eDRAM-LUT-Based CIM

The key challenge to build an efficient LUT-based digital CIM macro is the high-density memory cell design. Only when the inflated weight array has a smaller area than corresponding accumulation circuits, can the LUT-based optimization bring benefits. However, previous eDRAM-based CIM work typically employed thick oxide transistors or extra capacitors at the storage node to improve the retention time. The resulted cell area, though having less transistors, was even larger than an SRAM cell at the same technology node [30], [33]. Such solutions sacrificing the storage density for refreshing overhead are incompatible with the principle of LUT-based computing. As shown in Fig. 2(a), we adopt a compact eDRAM cell layout without additional MOM capacitors. As a result, the retention time is degraded by  $10\times$ . In such case,

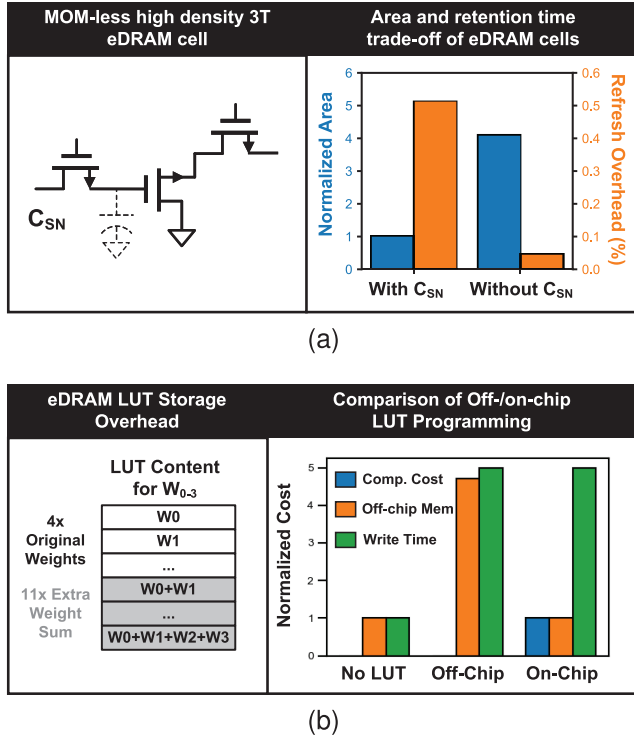


Fig. 2. Main challenges of eDRAM-LUT-based digital CIM. (a) Refreshing overhead of MOM-less high-density eDRAM and (b) LUT programming overhead.

frequent periodic refreshing operation will incur more than 50% overhead for computation operations at the worst scenario with high temperature and low operating frequency.

Besides the eDRAM issue, the LUT itself also introduces an additional programming cost, since both original weights and their linear combination need to be stored. Fig. 2(b) compares two typical solutions: off-chip and on-chip LUT encoding. While first one avoids the weight generation overhead by precomputing the LUT content during compilation stage, its drawback is also obvious since both off-chip storage and programming time inflates by  $5\times$ . As a tradeoff between on- and off-chip resources, the on-chip encoding approach locally accumulates the weight summation with additional computation units. However, both approaches still need to load weights into CIM macro serially, which inevitably consumes more programming time limited by I/O bandwidth.

As a summary, implementing a compact memory cell and reducing the encoding cost are essential to make eDRAM-LUT-based CIM practical and efficient.

### III. RECONFIGURABLE eDRAM-LUT-BASED DIGITAL CIM MACRO ARCHITECTURE

The overall architecture of the proposed eDRAM-LUT-based digital CIM macro is shown in Fig. 3(a), highlighting three key innovations. First of all, the design consists of a 40-Kb reconfigurable eDRAM array that serves dual purposes for both computing and storage. Unlike the digital CIM macros relying solely on logic circuits for computation [11], [22], [34], the proposed LUT-based implementation achieves the storage density higher than a logic-rule SRAM [35] in memory

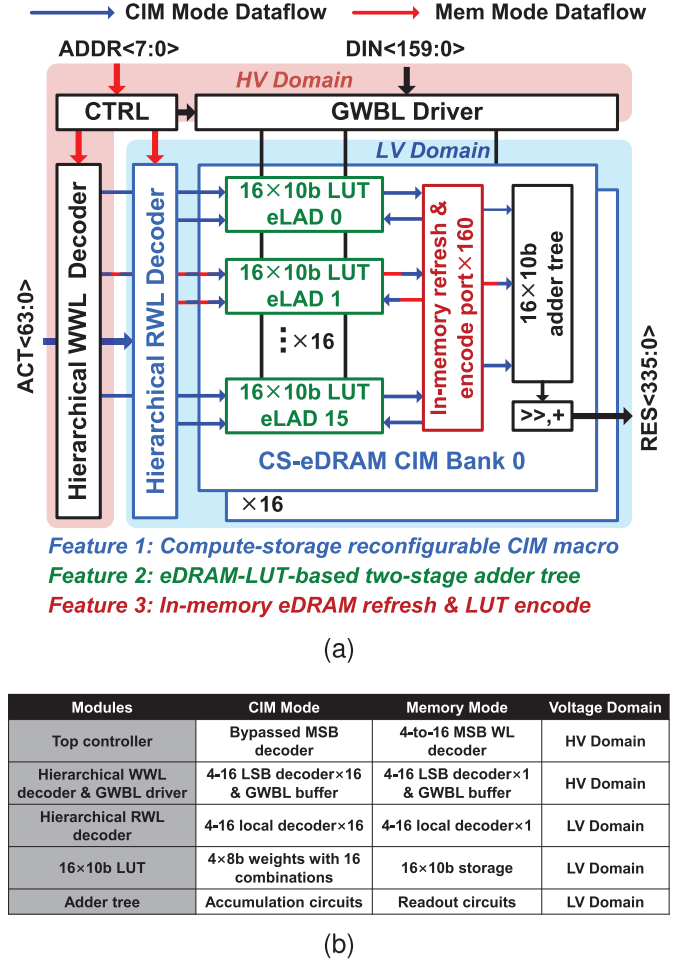


Fig. 3. (a) Overall architecture of the proposed eDRAM CIM macro with three key features and (b) function mapping of main components for CIM and normal memory modes.

mode while maintaining full input and output parallelism for high-throughput CIM operations. The whole eDRAM array is partitioned into 16 banks corresponding to different output channels, amortizing the WL decoder costs of 64-bit inputs.

The differences between CIM and normal memory mode mainly lie in the decoder configuration and LUT content. As shown in Fig. 3(b), the decoder is decomposed into LSB and MSB parts. In the memory mode, the MSB decoder is activated to select one of the 16 LSB decoders based on  $ADDR<7:4>$ , while the remaining address bits are forwarded to the selected LSB decoder to activate the corresponding WL. In contrast, the CIM mode bypasses the MSB decoder and enables all 16 LSB decoders, allowing for parallel LUT-based computation. The activation bits are directly used to fetch the corresponding partial sums. Consequently, the LUT content in CIM mode consists of the 16 combinations of  $4 \times 8$  b weights instead of the  $16 \times 10$  b arbitrary data in the memory mode. The detailed operation of both CIM and memory modes is illustrated later in Section IV-B.

Second, each CIM bank incorporates of 16 eLAD modules to perform efficient MAC operations. The eLAD is equivalent

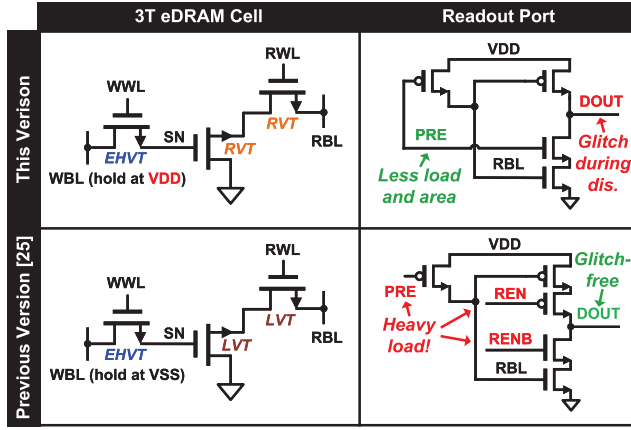


Fig. 4. Schematic of 3T eDRAM cells in this work and previous version [25] with different biasing schemes and readout port designs.

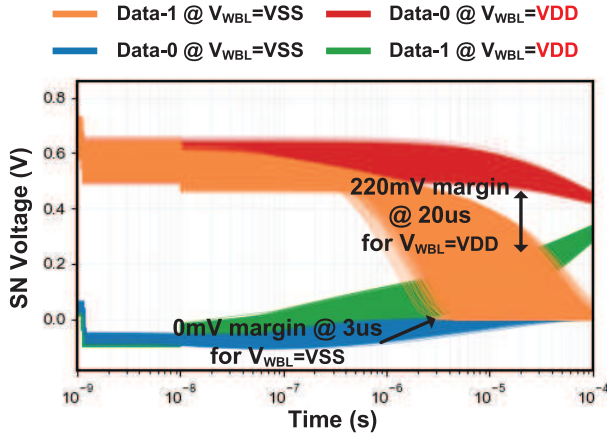
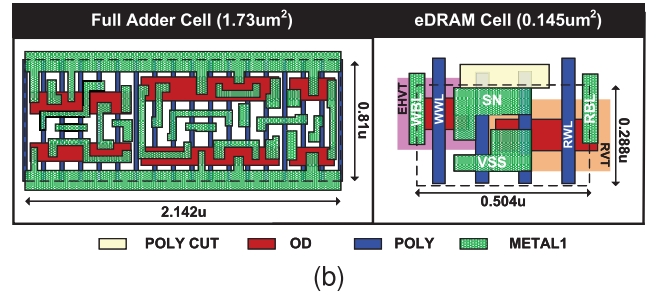
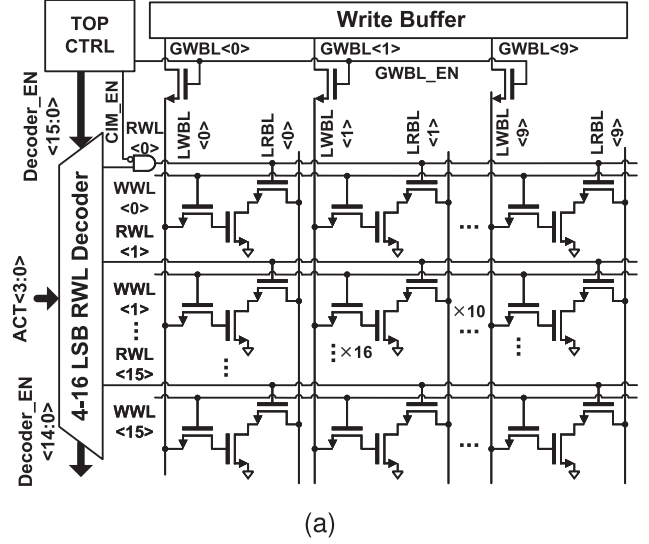


Fig. 5. 10k-point Monte Carlo simulation of leakage test at 60 °C and different WBL bias voltages.

to the first two stages of an adder tree, generating the MAC results of four input channels. Compared to the conventional digital logic implementations, the eLAD replaces complex full adders with simple 3 T eDRAM cells. Additionally, the dynamic power consumption caused by frequent switching is reduced through low-power memory read operations, thereby enabling both area- and energy-efficient MAC operations, as illustrated later in Section IV-B. The outputs of eLAD modules are subsequently aggregated through a  $16 \times 10$ -bit adder tree. To minimize the digital logic area, the CIM macro adopts bit-serial processing for input activations. Thus, the shift adders are employed to accumulate the partial sums in an LSB-first order over eight cycles, generating full-precision results for 64-input 8-bit MAC operations.

Lastly, dedicated local read and write ports are placed between eLAD modules and the following accumulation circuits. The bidirectional ports enable flexible and parallel data movements within the eDRAM array, an essential feature for efficient in-memory refreshing and weight encoding operations, as detailed in Section V. The macro also includes other peripherals for data movement and controlling, such as the I/O interfaces, timing modules, and the driver circuits for



| WL | Data       | WL | Data          | WL | Data          | WL | Data              |
|----|------------|----|---------------|----|---------------|----|-------------------|
| 0  | Don't Care | 4  | $W_3$         | 8  | $W_4$         | 12 | $W_3+W_4$         |
| 1  | $W_1$      | 5  | $W_1+W_3$     | 9  | $W_1+W_4$     | 13 | $W_1+W_3+W_4$     |
| 2  | $W_2$      | 6  | $W_2+W_3$     | 10 | $W_2+W_3$     | 14 | $W_2+W_3+W_4$     |
| 3  | $W_1+W_2$  | 7  | $W_1+W_2+W_3$ | 11 | $W_1+W_2+W_4$ | 15 | $W_1+W_2+W_3+W_4$ |

Fig. 6. (a) Schematic of eLADs, (b) layout implementation of the 3 T eDRAM cell compared to a standard nine-track full adder cell, and (c) data mapping of eLAD in CIM mode.

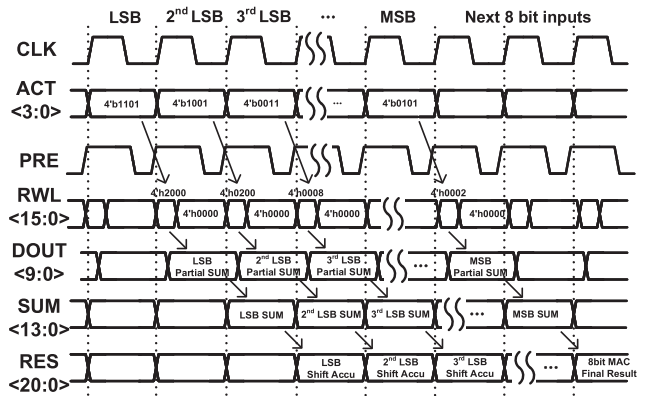


Fig. 7. Timing diagram of eLAD-based CIM operations.

global signals. Notably, different from [25], the write circuits, including WWL decoders, GWBL, and WWL drivers, are grouped in an isolated high-voltage domain, separating them



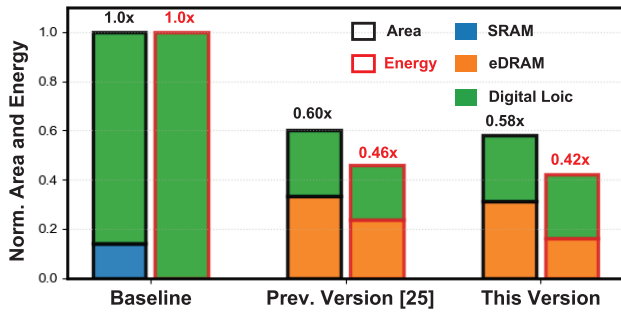


Fig. 8. Area and energy improvement of eLADs over the baseline digital CIM implementation.

from the other low-voltage computing circuits and the eDRAM array. Such isolation can mitigate the threshold voltage loss in write selectors of eDRAM cells, especially in low-voltage operating conditions, as shown in Section VI.

#### IV. ELAD TREE IMPLEMENTATIONS

##### A. Design Details of Optimized 3T eDRAM Cell

Before discussing the design tradeoffs of eLAD, we first elaborate on the implementation details and modifications of the 3 T eDRAM cell compared to our previous work as shown in Fig. 4 [25]. As outlined in Section II-B, the removal of additional capacitors at the storage node enables a more compact layout. To compensate for the resulting reduction in retention time without compromising area efficiency, we adopt an alternative approach to suppress the leakage. Rather than employing negative WWL voltages, which requires dedicated negative voltage generator [36], we modify WBL bias voltage from to VSS to VDD to naturally establish a negative  $V_{GS}$  for bit-“0.”

Fig. 5 presents the Monte Carlo simulation results comparing leakage behavior under different configurations. When the WBL is biased at VSS, the bit-“1” rapidly decays to VSS within 3  $\mu$ s at the fastest corner, thereby limiting the overall retention time of the eDRAM macro. In contrast, the leakage current can be greatly reduced for bit-“1” by biasing the WBL at VDD. Although this configuration causes bit-“0” to be charged toward bit-“1” gradually, the increased storage node voltage introduces a negative  $V_{GS}$ , which effectively suppresses the channel leakage current. To further mitigate the impact of increased bit-“0” voltage, the pair of read transistors in the eDRAM cell is changed from LVT to RVT devices. As a result, 220-mV voltage margin is preserved at 20  $\mu$ s, leading to a more than 7 $\times$  improvement in retention time, as presented in Section VI.

Additionally, the readout port design has been simplified to further reduce the area and energy consumption of the eDRAM array, as illustrated in the right column of Fig. 4. The previous design employed a C<sup>2</sup>MOS-like dynamic latch to eliminate potential glitches during RBL precharging and discharging phases, at the cost of driving additional complementary control signals REN and RENB. The current implementation, similar to a TSPC register [37], employs a single NMOS transistor instead of a transmission gate while sharing the PRE signal.

Although the modification introduces output glitches during RBL discharging, the total power and area are reduced by 8% and 3% due to simplified readout port design.

##### B. Efficient eLAD

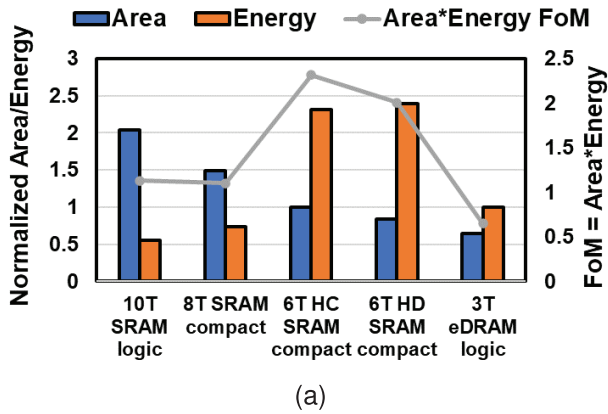
Fig. 6(a) presents the implementation of the proposed LUT-based adder that utilizes compact 3 T eDRAM cells. The design incorporates a  $16 \times 10$ -bit eDRAM subarray with hierarchical write bitlines and the local RWL decoder shared by different CIM banks. In the CIM mode, the eLAD performs MAC operations of  $4 \times 1$ -bit inputs and  $4 \times 8$ -bit weights per cycle through LUT access. As demonstrated in Fig. 6(b), the high-density memory cells enable efficient LUT-based computing, with the implemented eDRAM cell occupying merely 0.145  $\mu$ m<sup>2</sup>, which is 11.9 $\times$  smaller than that of a standard nine-track full adder cell.

Fig. 6 illustrates the weight data mapping organized by corresponding activation binary codes. This configuration allows the input bits to be directly used as LUT access address, thereby making the decoder logic consist with normal memory operation except for all-zero inputs. Since such case produces zero outputs irrespective of stored weights, the corresponding wordline is power-gated in the CIM mode by turning off discharging. Although it appears to waste storage capacity, the additional wordline can both simplify the decoder logic in memory mode and serve as a scratchpad for in-memory encoding operation, as discussed in Section V.

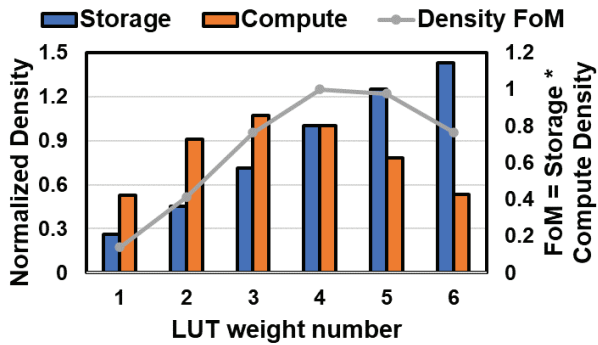
Fig. 7 shows the detailed timing diagram of the eLAD and subsequent accumulation circuits for an 8-bit MAC operation. The entire process follows a three-stage pipeline: 1) eLAD partial sum generation; 2) adder tree accumulation; and 3) bit-shift accumulation. First, the input activations are fed to the CIM macro in a bit-serial, LSB-first manner. The precharge signal initializes the RBL to prepare for readout. Then, for every four bits of activation data, a corresponding RWL pulse is generated to retrieve the partial sum DOUT from the eDRAM LUT. Second, these partial sums are accumulated by the adder tree, with its output SUM sampled on the rising edge of the next precharge pulse. Finally, the complete MAC result RES is generated by the shift-adders over eight cycles.

Compared to the baseline digital CIM composed of the standard digital cell and adopting the same input bit-serial architecture, as shown in Fig. 8, the proposed eLAD-based design achieves 40% area and 54% energy reduction while maintaining the same throughput and full precision. The improvements come from both the high-density memory cell and efficient LUT read operations. The current version achieves further reduction by simplifying the readout port, as illustrated in Fig. 4. In addition to the efficiency improvement, the proposed LUT-based computing strikes a balance of storage and computing. The eDRAM array dominates the overall area, leading to significant storage density in the memory mode.

Fig. 9 further explores the potential design space of LUT-based computing. First, we compare different LUT implementations using SRAM cells ranging from 10 to 6 T, which reveals a tradeoff between area and readout energy, as shown in Fig. 9(a). When accounting for dummy cells



(a)



(b)

Fig. 9. Design space exploration of LUT-based adders on (a) memory cell and (b) weight number selection.

surrounding the  $16 \times 10$ -bit array, the 3 T eDRAM achieves a 36% and 22% area reduction compared to foundry-provided 6 T high-performance and high-density SRAM, respectively. Although the 10 T SRAM [38], featuring a precharging-free and single-ended readout port, achieves the highest energy efficiency, the additional transistors result in a  $3\times$  area overhead, making it impractical for area-sensitive LUT-based computing. By adopting the area-energy product as a figure of merit (FoM), the FoM curve demonstrates that the proposed eDRAM-based solution achieves the optimal tradeoff of area and energy efficiency.

In addition to memory cell selection, Fig. 9(b) analyzes the impact of different LUT depths. As expected, the number of weights results in an exponential growth in memory size. While increasing the storage ratio and memory density, the overall benefit of LUT-based computing diminishes gradually. Specifically, when  $>5$  weights are contained, the area efficiency of eLAD drops below that of standard digital logic. By jointly considering storage and computational density, the FoM curve defined as their product reveals that the adopted four-weight LUT achieves the best tradeoff in density.

## V. IN-MEMORY REFRESHING AND ENCODING SCHEMES

### A. In-Memory Refreshing Implementations

After analyzing the advantages and tradeoffs of eDRAM-LUT-based CIM, we introduce how to overcome its refreshing

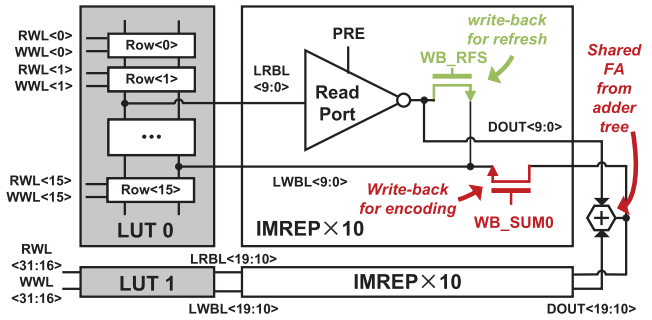


Fig. 10. Schematic of in-memory refreshing and encoding port.

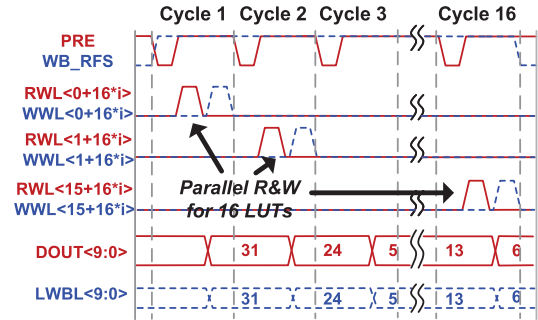


Fig. 11. Illustration of in-memory refreshing operations with detailed waveforms.

and programming overhead using the in-memory refreshing and encoding port. The proposed architecture contains a read port, which has been illustrated in Fig. 10, and two additional transistors marked in green and red, respectively.

The green transistor, controlled by the refresh write-back signal (WB\_RFS), enables in situ refreshing through connecting the output of read port to LWBL. The red transistor, in contrast, redirects summation results from the following adder tree to LWBL for generating different weight combinations.

The detailed waveforms of in-memory refreshing are illustrated in Fig. 11. The timing of refreshing operations can be divided into three phases: precharging, discharging, and write-back. The WB\_RFS signals are enabled following RWL pulse to realize write-back operation in one cycle. Unlike conventional row-by-row refreshing scheme, the in-memory refreshing activates different LUTs in parallel to achieve a  $16\times$  speedup. As shown in Section VI, the accelerated refreshing relaxes the constraint on eDRAM retention time.

### B. In-Memory Encoding Implementations

The in-memory encoding operation is similar to the refreshing one with a main difference that different weight combinations are written back, instead of the original readout data. In order to minimize the area overhead of weight computing circuits, we reuse the adders from the subsequent adder tree. For the compatibility with the adder tree dataflow, where inputs are read from two neighboring LUTs, the initial weights require a double writing scheme.

The whole encoding process is illustrated in Fig. 12, which are divided into two stages: the weight loading and generation.

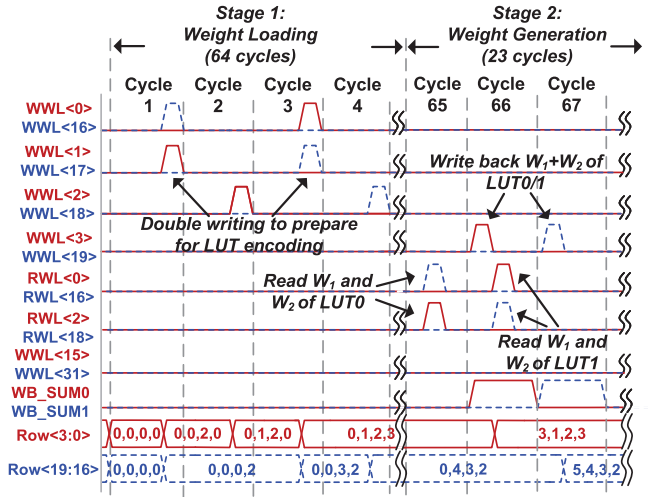


Fig. 12. Illustration of in-memory encoding operations with detailed waveforms.

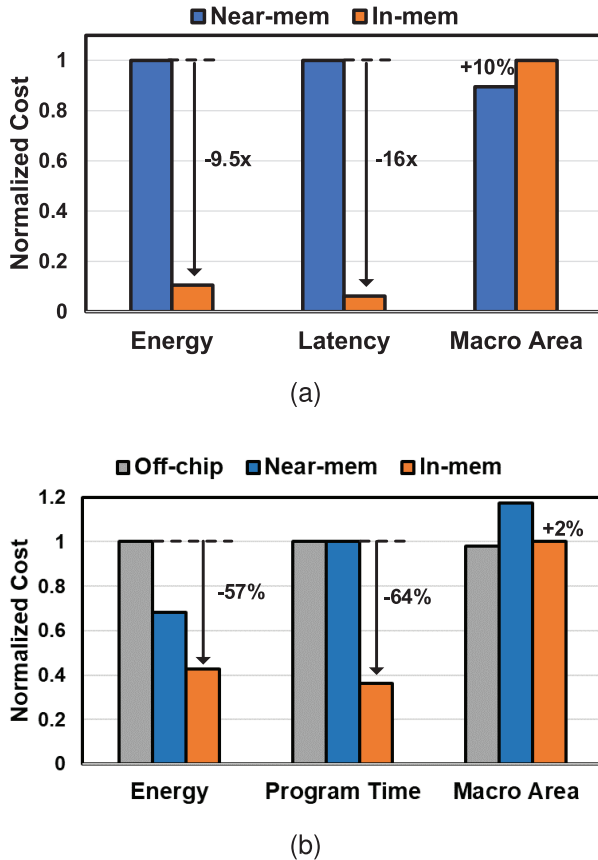


Fig. 13. Latency and efficiency improvements of in-memory (a) refreshing operations and (b) encoding operations.

Each weight data is stored in the corresponding LUT as well as its neighboring LUT to prepare for the next generation stage. The unused wordlines including WL0 serve as an intermediate weight buffer. Such double writing operation can be performed without latency overhead through activating both WWL in the same cycle.

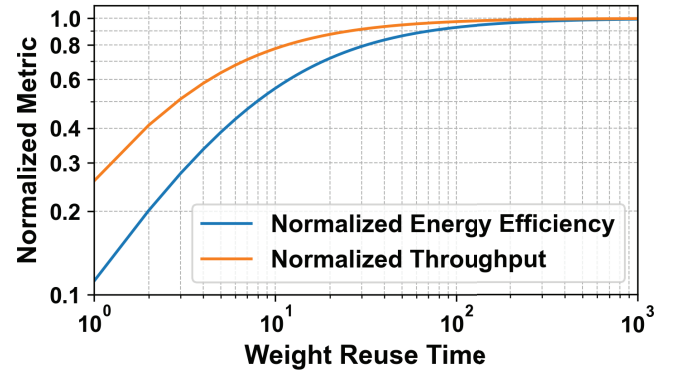


Fig. 14. Normalized macro energy efficiency and throughput considering the data loading time and energy under different weight reuse times.

During the generation process, the shared adder fetches intermediate data from two LUTs and write back the computation results by activating WB\_SUM signal. To ensure sufficient setup time for weight computing, the write back signal, unlike the refreshing mode, is aligned with the precharging signal in the next cycle. Due to the parallel encoding, the weight generation stage takes up 23 cycles, 26% of the total programming time. Fig. 13 shows the whole process of in-memory encoding, where 1st, 7th, 11th, and 13th rows of each LUT are used to store the temporary value for LUT generation. By carefully arranging the order of weight read and write operations and leveraging the unused WL0, the whole process is data-collision free. Also, since the encoding stage requires no data interaction with external logic, it allows the LUT-based CIM macro to maintain a programming model similar to regular memory.

Fig. 14 summarizes the in-memory refreshing and encoding schemes by the evaluation results. Leveraging the parallel in-memory bandwidth, the proposed design achieves 16 $\times$  and 2.9 $\times$  reduction in refreshing and programming latency, respectively. In addition to the speed up, the in-memory technique improves 9.5 $\times$  and 2.3 $\times$  energy efficiency by reducing global data movement. The additional local read-write ports and write-back path from the adders increase the macro area by only 10% and 2%, respectively.

We also evaluate the normalized energy efficiency and throughput of the LUT-based CIM macro considering the programming overhead under different weight reuse times, as show in Fig. 15. Since the in-memory encoding stage consumes more power than normal computing due to concurrent read and write operations, it requires more computing cycles to amortize the energy overhead than that of throughput. Nevertheless, when the weight reuse exceeds 70 $\times$ , which is typical in CNN models, the programming overhead accounts for less than 10%.

## VI. MEASUREMENT RESULTS

Fig. 16 presents the die photograph of the proposed eDRAM CIM macro and summarizes its main measurement results. The prototype chip is fabricated using 28-nm HKMG technology. The eDRAM macro occupies a 0.016 mm<sup>2</sup> area with a 40-Kb

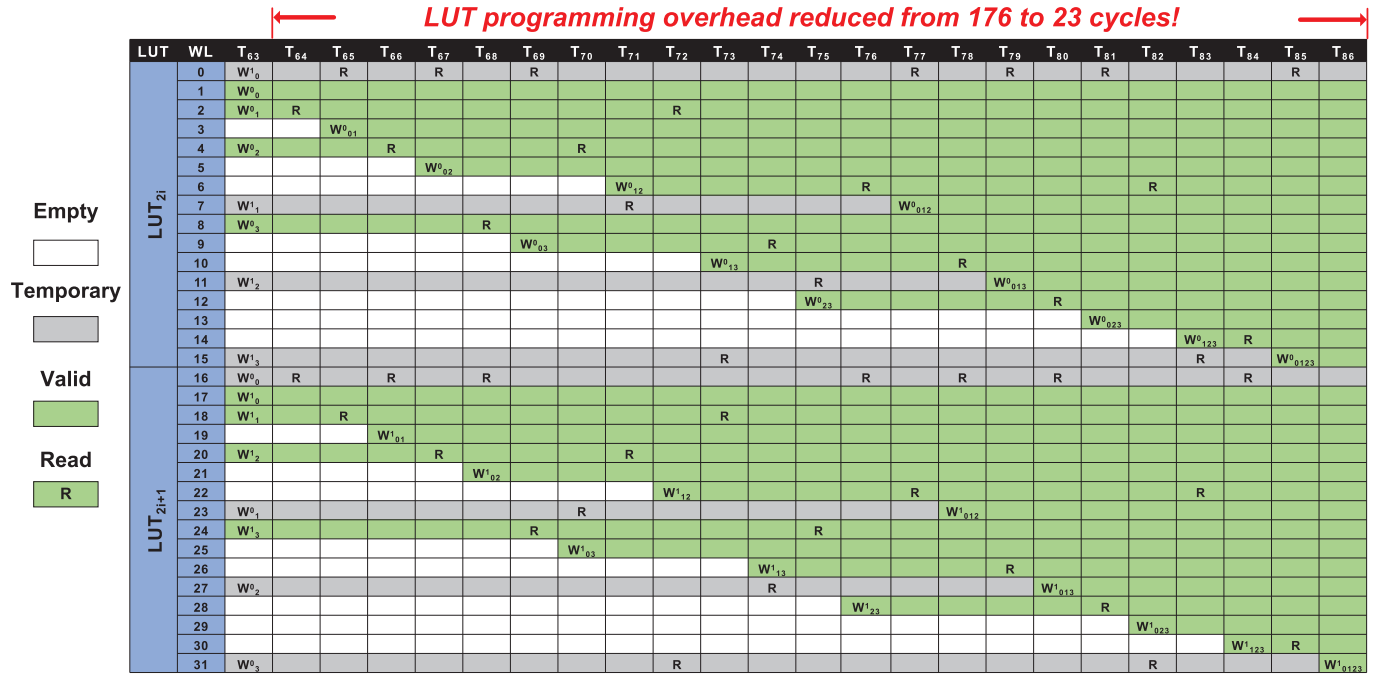


Fig. 15. Whole process of in-memory encoding for two neighboring LUTs in 23 cycles. The superscript of  $W$  denotes the number of the LUT, while the subscript denotes the combination of weights.

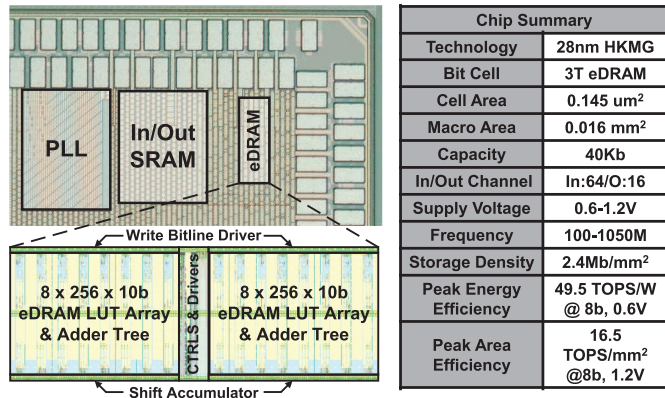


Fig. 16. Die photograph of fabricated prototype and chip summary.

array. The test chip also includes a PLL for peak frequency test and SRAM buffers for input and output data. Fig. 17 shows the testing setup, which includes the wire-bonded IC, the custom PCB, and an FPGA serving as a bridge between the chip and host PC. A thermoelectric heat pump controls the chip temperature, which is monitored by the thermal imager for retention test.

Fig. 18(a) demonstrates the voltage–frequency scaling characteristics. The separated voltage domains for write driver and computing circuits enable CIM operation at reduced voltages down to 0.6 V with 100-MHz frequency. The peak performance of 1.05 GHz is obtained at 1.2 V. Although the design exhibits minor performance degradation at higher voltages, compared to [25], due to limited discharging speed of RVT read transistors, its area efficiency is maintained due to optimized layout.

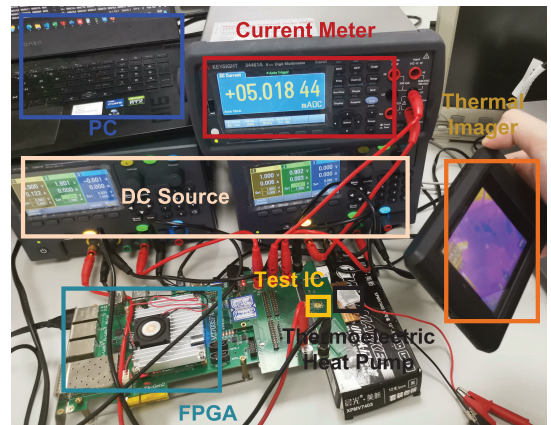
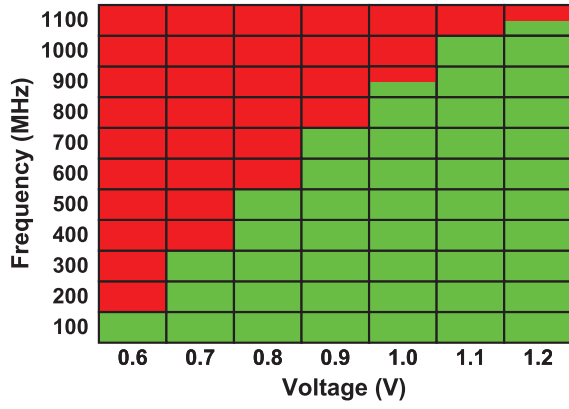


Fig. 17. Measurement setup.

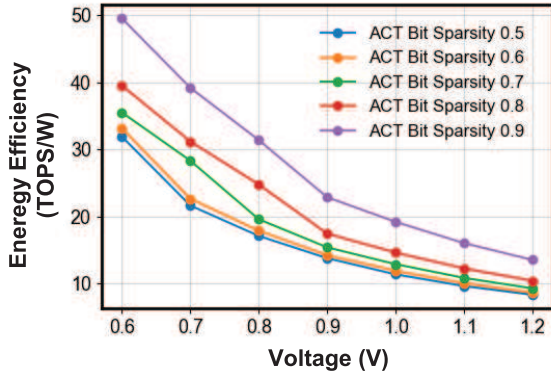
Fig. 18(b) presents the 8-bit energy-efficiency measurements across varying supply voltages and input bit sparsity levels, with weight bit sparsity fixed at 50%. A peak energy efficiency of 49.5 TOPS/W is achieved when operating at 0.6 V with 90% input sparsity.

Note that the voltages shown in Fig. 18 and subsequent Fig. 19 refer to the low-voltage domain of the computing circuits and the eDRAM array, which account for more than 80% of total power consumption, as indicated in Fig. 18. The high-voltage domain, which comprises the eDRAM write controllers and drivers, operates at 1.1 V when computing voltage is below 1.1 V and at 1.2 V otherwise.





(a)



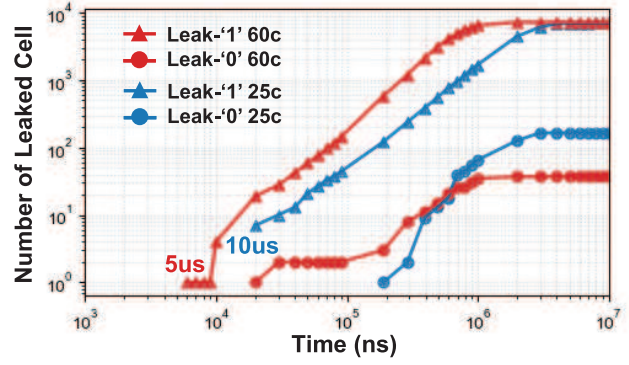
(b)

Fig. 18. (a) Shmoo plot and (b) voltage-efficiency scaling result.

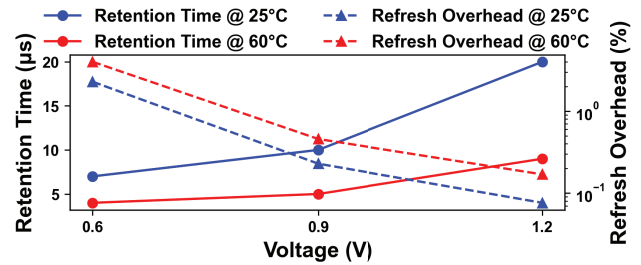
Fig. 19(a) presents the number of leaked eDRAM cell changing over time under varying temperatures at nominal voltage. The leaked cells have a different saturated number under various conditions due to our optimized bias scheme. As shown in the Monte Carlo simulation in Fig. 5, the final state of eDRAM cell lies between “1” and “0” as the result of balanced leakage paths to both WBL at VDD and substrate at VSS. For most cells, the balanced voltage is lower than the threshold of “1,” causing them gradually decay to “0” over time. However, for some bits, the balanced states are still recognized at “1.” Such balanced states are sensitive to PVT conditions, resulting in a varied distribution across different corners.

Fig. 19(b) further shows the retention time across different voltage and temperature corners. While the retention time is degraded to  $4 \mu\text{s}$  at the worst case of 0.6 V and  $60^\circ\text{C}$ , the eDRAM CIM macro maintains a refresh overhead less than 5%, due to the in-memory refresh technique.

Fig. 20 illustrates the detailed breakdown results. Compared to prior digital-dominated CIM designs, the proposed architecture demonstrates a more balanced computing-storage ratio. The bottom of the figure shows the layout of an eDRAM CIM bank employing a sandwich-like floorplan, where adder trees and shift-accumulation circuits occupy less than half area. For the energy breakdown, the adder tree accounts for 49% of the total power consumption, which is increased compared



(a)



(b)

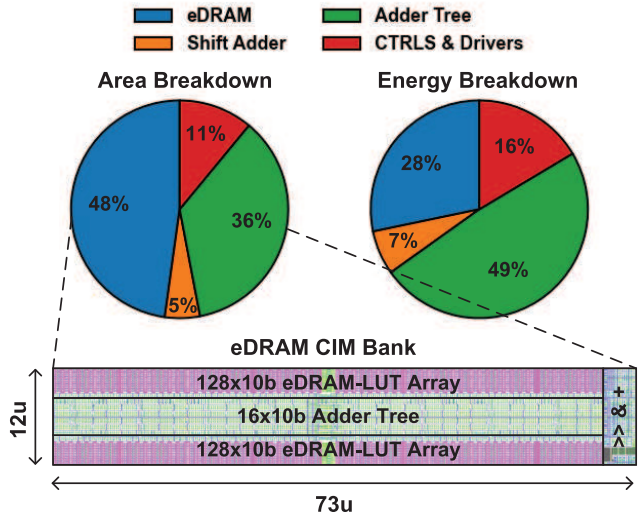
Fig. 19. (a) Number of leaked cells at 0.9-V voltage and  $25^\circ\text{C}$ – $60^\circ\text{C}$  and (b) retention time and refreshing overhead of different voltage and temperature corners.

Fig. 20. Area and energy breakdown of the eDRAM-LUT-based CIM macro.

with our previous implementation [25], due to glitches at eDRAM-LUT outputs. However, the overall energy efficiency is improved due to simplified eDRAM readout circuits and control signals.

Table I compares this work with prior eDRAM- and SRAM-based analog and digital CIM designs. The proposed design combines the benefits of eDRAM and digital CIM approaches, offering both high density and accuracy. Compared to previous analog CIM designs, which suffered from the efficiency and

TABLE I  
COMPARISON WITH PREVIOUS eDRAM AND SRAM CIM MACROS

|                                        | VLSI'23 C10-3 | ISSCC'23 16.5      | ISSCC'23 7.8           | This Work                      | ISSCC'24 34.4         | ISSCC'22 16.1                   | VLSI'22 C2.5            |
|----------------------------------------|---------------|--------------------|------------------------|--------------------------------|-----------------------|---------------------------------|-------------------------|
| Technology                             | 28nm          | 28nm               | 22nm                   | 28nm                           | 3nm                   | 28nm                            | 12nm                    |
| MAC Operation                          | Analog Charge | Analog Charge      | Analog Charge          | Digital Memory-LUT             | Digital Adder tree    | Digital Approx.                 | Digital Logic-LUT       |
| Bit Cell                               | 3T eDRAM      | 4T eDRAM           | 9T SRAM                | 3T eDRAM                       | 6T SRAM               | 8T SRAM                         | 6T SRAM                 |
| Macro Capacity                         | 100Kb         | 100Kb              | 64Kb                   | 8/40Kb (CIM/Mem)               | 60.75Kb               | 16Kb                            | 8Kb                     |
| Macro Area (mm <sup>2</sup> )          | 0.097         | 0.103              | 0.21                   | 0.016                          | 0.0157                | 0.033                           | 0.0323                  |
| Storage Density (Mb/mm <sup>2</sup> )  | 1.0           | 0.95               | 0.30                   | 0.48/2.4 (CIM/Mem)             | 3.78                  | 0.50                            | 0.25                    |
| # of In/Out Ch                         | 64 / 32       | 320 / 64           | 64 / 256               | 64 / 16                        | 72 / 4                | 256 / 64                        | 64 / 16                 |
| Input / Weight / Output Bits           | 1:8 W:9       | 1: 5/9 W: 4/8 O:13 | 1:8 W:8 O:22           | 1:1-8 W:8 O:22                 | 1:12 W:12             | 1: 1 W: 1 O:7                   | 1: 4-8 W: 4/8 O:18      |
| 8b* Energy Effi. (TOPS/W)              | 45.7 (@1.0V)  | 30.1 (@1V)         | 21.4/18.5 (@0.72/0.8V) | 49.5/22.9/13.5 (@0.6/0.9/1.2V) | 121 (@0.55V)          | 34.7/17.3/ 14.5 (@0.5/0.9/1.1V) | 30.3 (@0.72V)           |
| 8b* Area Effi. (TOPS/mm <sup>2</sup> ) | 0.81 (@1V)    | 2.03 (@1V)         | 1.19/1.44 (@0.72/0.8V) | 1.57/11.0/16.5 (@0.6/0.9/1.2V) | 66/123.8 (@0.55/0.9V) | 4.3/9.5 (@0.9/1.1V)             | 10.4/14.3 (@0.72/0.92V) |
| Full Precision ?                       | No            | No                 | Yes                    | Yes                            | Yes                   | No                              | Yes                     |
| Normalized** Density FoM***            | 0.81          | 1.92               | 0.36                   | 39.6                           | 6.8                   | 4.75                            | 0.88                    |
| Normalized** Efficiency FoM****        | 37.0 (@1V)    | 61.1 (@1V)         | 21.7 (@0.8V)           | 251.9 (@0.9V)                  | 218.4 (@0.55V)        | 75.1 (@0.9V)                    | 107.0 (@0.72V)          |

\*Normalized to 8bit operations by 1xW bits \*\* The area efficiency, energy efficiency and storage density are all normalized to 28nm based on the foundry SRAM bitcell size.

\*\*\* Density FoM = Peak Area Efficiency x Storage Density in memory mode \*\*\*\* Efficiency FoM = Energy x Area Efficiency @ Nominal voltage

TABLE II  
NEURAL NETWORK PERFORMANCE AND EFFICIENCY BENCHMARKS

| Neural Network Model                                     | ResNet-50   | ResNet-18 |
|----------------------------------------------------------|-------------|-----------|
| Dataset                                                  | ImageNet-1k |           |
| Bit Precision                                            | 8b/8b W/A   | 8b/8b W/A |
| Accuracy (%)                                             | 79.38       | 69.75     |
| Compute/Programming Energy per Class (uJ) <sup>1,2</sup> | 378.5/15.6  | 194.6/7.6 |
| Throughput (fps) <sup>1,2</sup>                          | 22.2        | 48.3      |

<sup>1</sup>Only consider macro programming, computing and refreshing energy/latency for batch size of 1 <sup>2</sup>Evaluated at 0.9V/700MHz

SNR tradeoff, this work achieves higher throughput and energy efficiency without compromising accuracy. Compared to other digital CIM implementations, including standard, approximate, and logic-LUT-based approaches, the proposed eDRAM-LUT-based design achieves the highest area and energy efficiency product at nominal voltage when normalized to 28-nm technology node. Besides, the proposed LUT-based design achieves a superior storage density of 2.4 Mb/mm<sup>2</sup> in the memory mode, leading to a 5.8× normalized density FoM. Although the effective density is reduced by 5× in CIM mode due to expanded LUT storage, it remains 1.9× higher than that of a 12-nm logic-LUT-based design.

Table II presents the benchmarking results of the proposed eDRAM-LUT-based macro on typical neural network models. It achieves 215.2/394.1 μJ per image energy consumption and 48.3/22.2 frames/s throughput for ResNet-18/ResNet-50 models on ImageNet dataset. Due to the efficient in-memory refreshing and encoding techniques, the total overhead of LUT programming and eDRAM refreshing is below 5%.

## VII. CONCLUSION

Digital-based approach is a promising candidate to implement reliable and large-scale CIM architecture, due to its exceeding performance, robustness, and scalability. However, the existing digital CIM solutions mainly focus on optimizing computing logic and dataflow, instead of taking memory and computing circuits as a whole.

The proposed eDRAM-LUT-based design can break the limitation of digital logic and combine the benefits of high-density eDRAM cell and high-throughput digital CIM architecture. The power and area consuming digital datapath are replaced by efficient LUT. The eDRAM-LUT-based implementation achieves 42% and 58% area and energy reduction compared to traditional digital design methodology. To overcome the eDRAM leakage and LUT programming issues, the in-memory refreshing and weight encoding techniques are developed. Leveraging the parallel in-memory bandwidth, the refreshing and programming latency are reduced by 16× and 2.9×.

A 28-nm prototype is fabricated and measured to validate the proposed eDRAM-LUT-based CIM design. The measured results show a peak 8-bit energy and area efficiency of 49.5 TOPS/W and 16.5 TOPS/mm<sup>2</sup> with 2.4-Mb/mm<sup>2</sup> storage density in the memory mode. The efficiency FoM and density FoM surpass previous work by 1.2× and 5.8×.

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